

Demo Day Runbook — RAG Chatbot

Before you leave (do this today, on your current network)

1. Run the full 3-terminal sequence below, start to finish, on this network.
 2. Confirm a real question gets a real answer in the browser.
 3. **Do not delete** `venv`, `node_modules`, or the `rag_chatbot` sibling folder (the one with the downloaded embedding model) between now and the demo. Everything that needed internet access has already happened — your job at DRDO is just to *start* already-installed software, not install anything new.
 4. If DRDO's network blocks outbound traffic (common at research/government facilities), this is what makes your demo network-independent. Don't risk it by "cleaning up" beforehand.
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The actual startup sequence (3 terminals, every time)

Terminal 1 — FastAPI (RAG engine)

```
powershell

cd path\to\rag_chatbot_v1-master
.\venv\Scripts\Activate.ps1
uvicorn api:app --reload --port 8000
```

Wait for: `Loaded index with N vectors` — confirms the FAISS index loaded correctly.

Terminal 2 — Express backend

```
powershell

cd path\to\backend
npm start
```

Wait for: `PostgreSQL Connected` and `Server running on port 5000`.

Terminal 3 — Frontend

```
powershell

cd path\to\frontend
npm run dev
```

Wait for: a local URL printed, usually `http://localhost:5173`.

Then in the browser:

1. Open the frontend URL
 2. Register or log in
 3. Go to the Query page, type a real question, click Ask
 4. First question after a fresh `uvicorn` start will be slower (TinyLlama loading into memory) — this is normal, don't panic or restart anything
-

Quick pre-flight checks (run these first if anything seems off)

```
powershell

# Is Postgres actually running and reachable?
psql -U postgres -d drdo_assistant -c "\dt"
```

Should list `users`, `documents`, `chat_history`, `sessions`. If `psql` isn't recognized, see the fix below.

```
powershell

# Is anything already squatting on the ports you need?
netstat -ano | findstr "8000"
netstat -ano | findstr "5000"
netstat -ano | findstr "5173"
```

If a port shows up unexpectedly occupied, something (often the old mock-rag-server) is still running. Find the PID at the end of that line, then:

```
powershell

taskkill /PID <the_number> /F
```

Known issues we already hit once — fixes on hand

`venv\Scripts\Activate.ps1` not recognized

```
powershell

& ".\venv\Scripts\Activate.ps1"
```

If that also fails, use the full absolute path instead of `.\`.

`psql` not recognized

```
powershell
```

```
& "C:\Program Files\PostgreSQL\<version>\bin\psql.exe" -U postgres -d drdo_assi
```

(check `dir "C:\Program Files\PostgreSQL"` for the exact version folder) Or just use pgAdmin's Query Tool instead — no PATH issues there.

`python3` not recognized on Windows Always use `python`, not `python3`, on this machine.

scikit-learn `ModuleNotFoundError: No module named 'sklearn.utils'`

powershell

```
Remove-Item -Recurse -Force ".\venv\Lib\site-packages\sklearn"  
Remove-Item -Recurse -Force ".\venv\Lib\site-packages\scikit_learn-*"  
pip install scikit-learn
```

`ValueError: Unrecognized model in ../rag_chatbot/my_embedding_model` The model folder is empty/corrupted — redownload:

powershell

```
Remove-Item -Recurse -Force "..\rag_chatbot\my_embedding_model"  
python download_model.py
```

`Error loading ../rag_chatbot/assets/sample_manual.pdf` / `"Embedded ==0=="` The sibling assets folder is missing the PDF:

powershell

```
mkdir ../rag_chatbot/assets  
copy assets/sample_manual.pdf ../rag_chatbot/assets/sample_manual.pdf
```

Server crashes / connection drops right when asking a question (likely out-of-memory)

Confirm `generator.py` uses `torch.bfloat16`, not `torch.float32`, in the `from_pretrained(...)` call. This was already fixed once — just confirm it's still that way.

Port already in use when starting uvicorn The mock `index.js` (or a previous uvicorn instance) is still running on 8000. Find and stop it (see port-check commands above), or just close that terminal entirely and open a fresh one.

401 "Invalid Token" / "No token provided" on the Query page You're not actually logged in, or `localStorage` lost the token (e.g., browser restarted, different browser/profile). Just log in again through the frontend before testing.

Worst case: different machine, nothing installed, no internet

If you somehow end up on a completely fresh machine with no internet at DRDO, **this**

likely cannot be set up from scratch on-site — pip, npm, and the model downloads all need internet. The realistic mitigation is: bring this exact laptop, already set up, as the demo machine — not a backup plan to reinstall there. If a fresh setup is genuinely unavoidable and there IS internet:

```
powershell

# Backend
cd backend
npm install

# Frontend
cd frontend
npm install

# RAG service
cd rag_chatbot_v1-master
python -m venv venv
.\venv\Scripts\Activate.ps1
pip install -r requirements.txt          # after removing the "pickle" line first
pip install fastapi "uvicorn[standard]" python-multipart
pip install torch --index-url https://download.pytorch.org/whl/cpu
pip install transformers accelerate

mkdir ..\rag_chatbot\my_embedding_model
python download_model.py

mkdir ..\rag_chatbot\assets
copy assets\sample_manual.pdf ..\rag_chatbot\assets\sample_manual.pdf

python main.py      # confirms the pipeline works before starting the API
```

Then proceed with the normal 3-terminal startup sequence above. Budget at least 20–30 minutes for this if it comes to it — mostly waiting on downloads (torch + TinyLlama are the big ones, ~2.5GB combined).